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Studierichting: Informatica
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$$15 \cdot 3^{x-1} - 6 \cdot 9^x = 1$$

$$\Leftrightarrow 15 \cdot 3^{x-1} - 6 \cdot 3^{2x} = 1$$

$$(s = 3^x)$$

$$\Leftrightarrow 15 \cdot \frac{s}{3} - 6 \cdot s^2 = 1$$

$$\Leftrightarrow 5s - 6s^2 = 1$$

$$\Leftrightarrow 6s^2 - 5s + 1 = 0$$

$$\Rightarrow s = \frac{5 \pm \sqrt{25 - 24}}{12} = \frac{5 \pm 1}{12}$$

$$s_1 = \frac{1}{2} \text{ \& } s_2 = \frac{1}{3}$$

$$(\text{Herleid terug naar } x : 3^x = s)$$

$$\left\{ \begin{array}{l} 3^x = \frac{1}{2} \Rightarrow x = \log_3 \left(\frac{1}{2} \right) \\ 3^x = \frac{1}{3} \Rightarrow x = \log_3 \left(\frac{1}{3} \right) = -1 \end{array} \right.$$

$$x_1 = \log_3 \left(\frac{1}{2} \right) \text{ \& } x_2 = -1$$

References